



Guideline

Venepuncture

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff

Introduction

Venepuncture is an invasive procedure used for diagnostic and monitoring purposes¹. It is the responsibility of the practitioner's involved with this procedure to ensure that each blood test is required to prevent unnecessary pain and distress for patients². Venepuncture is also often referred to as phlebotomy.

A valid medical order must be in place prior to initiation of venepuncture.

Risk Statement

Non-compliance with this will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☒
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☒
Other:	☐		

1.0 Safety Information

- 1.1. Standard precautions are to be maintained at all times during venepuncture, including the use of appropriate Personal Protective Equipment (PPE) sharps disposal within reach, and hand hygiene as per SCGOPHCG Policies.
- 1.2. The medical officer requesting the specimen is responsible for completing the request form or the Computerised Provider Order Entry (CPOE).
- 1.3. Venepuncture must be performed by, or under the direct supervision of, a practitioner who has successfully completed the appropriate training and competency assessment.¹
- 1.4. The practitioner must **not** exceed two (2) venepuncture attempts on any patient at any given time. ⁴ Assistance should be sought from a more experienced practitioner after 2 non-successful attempts by the initial collector. The subsequent collector must commence the procedure from Step 1.
- 1.5. Informed consent must be obtained from the patient prior to the procedure and also prior to a subsequent attempt by the same collector with an explanation provided about why reattempt is required.⁴
- 1.6. Single patient use tourniquets are to be used.⁵
- 1.7. Nurses are **not** to perform venepuncture on some categories of patients (e.g. renal patients). This will be determined at a specialty/departmental level.

2.0 Patient Identification

- 2.1. Prior to the procedure the practitioner must confirm with the patient their full name, date of birth and residential address.
- 2.2. Staff must check the patient's responses against the patient's ID band and/or other information provided in the patient's clinical record (as per SCGOPHCG Hospital Policy No. 48 Patient Identification and Procedure Matching).

3.0 Specimen Requirements

- 3.1. The person collecting the specimen is responsible for completing the collection details on the request form and labelling the specimen container prior to transport.
- 3.2. The patient's addressograph label should be used whenever available. Clearly label the specimen with:
 - Patients full name
 - Patients Unique Medical Record Number (UMRN)
 - Patients Date of Birth and
 - Date, time and site of collection.
- 3.3. If the specimen is urgent, please state clearly "URGENT" on request form.
- 3.4. Ensure specimen collection tubes have not exceeded their expiry date.
- 3.5. Fill each specimen collection tube as specified on the tube.
- 3.6. The specimen must be separated from the form in the dual compartment plastic bag, prior to transport. Place the specimen in the sealed compartment and send to PathWest as soon as possible after collection.
- 3.7. Ensure blood samples are collected in correct order as per PathWest recommended Order of Draw PSCP036.⁴
- 3.8. See [Appendix 1](#) for Specimen Collection Guide.

4.0 Sites for Venepuncture

- 4.1. Veins of the upper extremities, distal to and including the cubital fossa are the most suitable to use. The median cubital, cephalic or basilic veins are generally the best target¹. See [Figure 1](#).
- 4.2. Veins in the metacarpal area must only be used when the median cubital, cephalic or basilic veins are inaccessible.
- 4.3. Specimen collection should be avoided from the following sites:
 - Arms or hands receiving IV therapy.
 - Arms with working arterio-venous fistulas or grafts.
 - Badly bruised or thrombosed veins.
 - Recent catheter sites e.g. angiogram.
 - Arms or hands on the side of a mastectomy and/or axillary clearance, and/or arms with the presence of active lymphoedema.
 - Insides of wrists (due to numerous nerve endings).
 - Existing PIVC's.
 - Legs/feet. If this is not possible the procedure requires the foot to be elevated and the site monitored post collection.
 - An infected limb e.g. cellulitis.
- 4.4. In the instance that a sample is required from the sites above, confirm with medical staff prior to the procedure.

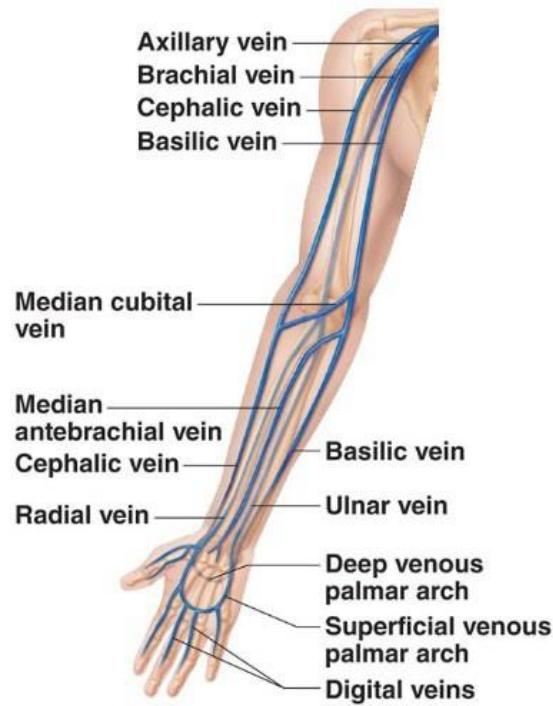


Figure 1: Veins of the upper extremity

5.0 Venepuncture Procedure

5.1 During the procedure

5.1.1. If the tourniquet is touched after donning gloves (e.g. for tightening and releasing purposes), all contact should be kept to a minimum and by using only the non-dominant hand.

5.1.2. If at any time during the collection procedure additional equipment is required which has not been assembled prior to the procedure, please ensure that gloves are removed, hand hygiene performed, and a fresh pair of gloves donned.

5.2 Approved venepuncture methods

Perform venepuncture according to one of the four approved blood collection methods outlined below:

- Hypodermic Safety needle with vacutainer
- Winged/Butterfly needle with vacutainer
- Hypodermic Safety needle with syringe
- Winged/Butterfly needle with syringe

The use of a blood collection set with a Vacutainer access device is best practice, providing a better sample and is safer, more convenient, and cost effective.

5.3 Venepuncture Procedure Equipment

- IV Trolley
- Absorbent sheet
- Small sterile dressing
- Adhesive Tape
- Personal Protective Equipment (PPE)
- 70% alcohol only wipe
- Sterile Gauze
- Tourniquet (disposable, single patient use) 4
- Vacutainer access device or blood transfer Vacutainer system and syringe

- Blood collection set (butterfly) or hypodermic safety needle (Recommended 21-23g)
- Blood collection tubes (As per PathWest 'Order of Draw PSCP036', available in the PathWest Pre Analytical Procedures Manual)
- Sharps disposal container
- Dual compartment biohazard bag

5.4 Procedural Steps

- 5.4.1. Confirm the patient identity using at least three identifiers (e.g. name, date of birth, residential address), explain the procedure and obtain verbal consent.
- 5.4.2. Check request form and test conditions.
- 5.4.3. Perform hand hygiene.
- 5.4.4. Gather and prepare equipment.
- 5.4.5. Perform hand hygiene.
- 5.4.6. Position patient and support limb, e.g. on a pillow with an absorbent sheet.
- 5.4.7. Apply tourniquet approximately 12cm above the venepuncture site; do not apply tourniquet for more than 2 minutes to prevent circulatory impairment and haemoconcentration.⁴
- 5.4.8. Perform hand hygiene and apply non sterile gloves.
- 5.4.9. Palpate and identify appropriate vein, checking for vein position and direction using one of the first two fingers on your non dominant hand (do not use your thumb).
- 5.4.10. Using a 70% alcohol only swab cleanse the site for 30 seconds, starting from the centre of the venepuncture site and work downwards and outwards to cover an area of 2 cm or more. Allow to dry completely (30 seconds).⁷
- 5.4.11. To ensure aseptic technique do not re-palpate. If necessary to re-palpate, repeat palpation and swabbing as per procedural steps 5.4.9 and 5.4.10.

5.5 Winged/Butterfly Needle with Vacutainer or Syringe

- 5.5.1. Complete steps 5.4.1 to 5.4.10
- 5.5.2. Remove the sheath from the needle (attached to either syringe or vacutainer).
- 5.5.3. Hold the ridged side of the butterfly wings with the tips of the thumb and index finger, tucking your remaining fingers into your palm, using your dominant hand.
- 5.5.4. Using your non dominant hand, apply traction 10-12cm's to the skin below the intended puncture site to anchor vein and prevent movement³.
- 5.5.5. With your dominant hand use a forward motion to insert the needle over and in-line with the vein, with the bevel facing up at a 15°- 30° angle to the skin⁴.
- 5.5.6. Insert needle using a steady advance and release traction once flashback has occurred⁴. Flashback will be evident once needle is in the vein. (i.e. flashback is when blood appears in the yellow chamber of the butterfly).
- 5.5.7. Stabilise the needle by securing the wings with tape.
- 5.5.8. Insert the collection tube fully into the vacutainer holder, so that the interior needle punctures the stopper of the tube. Leave in situ until filled as per minimum requirements. Refer to [Appendix 2: Venepuncture Troubleshooting](#) if flow is poor.
- 5.5.9. If using a syringe, pull back on the plunger with gentle pressure, to avoid haemolysis, using your free hand until a sufficient quantity of blood is obtained.
- 5.5.10. Continue procedure from Section 5.7.

5.6 Hypodermic Safety Needle with Vacutainer or Syringe

- 5.6.1. Complete procedural steps 5.4.1 to 5.4.10.
- 5.6.2. Remove sheath from needle (attached to either syringe or vacutainer).

- 5.6.3. Hold the needle holder in your dominant hand with the tips of the first two fingers underneath and your thumb on top, tucking your remaining fingers into your palm.
- 5.6.4. Using your non dominant hand, apply traction 10-12cms to the skin below the intended puncture site to anchor vein and prevent movement.³
- 5.6.5. With your dominant hand insert the needle over and in-line with the vein, with bevel facing up at a 15°- 30° angle to the skin.⁴
- 5.6.6. Insert needle and release traction.
- 5.6.7. Steady the needle:
 - a. Insert collection tube fully into the vacutainer holder, so that the interior needle punctures the stopper of the tube⁴- OR
 - b. Pullback on the plunger on the syringe with gentle pressure to avoid haemolysis, using your freehand until a sufficient quantity of blood has been obtained⁴.
- 5.6.8. Continue procedure from Section 5.7.

5.7 After the procedure

- 5.7.1. When blood withdrawal is complete, release tourniquet, withdraw needle and engage safety sheath.
- 5.7.2. Apply pressure to wound site using a clean gauze or cotton wool ball and apply a small non sterile dressing.
- 5.7.3. Dispose of sharps appropriately. If using a syringe, safely remove needle from syringe. Attach blood transfer device vacutainer and follow steps 5.5.8 and 5.5.9.
- 5.7.4. Invert tubes at least six times to ensure mixing of contents of tube with blood.⁴
- 5.7.5. Remove PPE and perform hand hygiene.
- 5.7.6. Label each blood tube with date and time of collection. Verify and initial labels on each specimen.
- 5.7.7. Where possible show the patient all samples and request form and ask the patient to confirm that the information on all labels is correct⁴
- 5.7.8. Place all samples and signed request form into a dual compartment biohazard bag.⁴
- 5.7.9. Check patients' venepuncture site for any sign of bleeding. Apply a clean dressing if required.
- 5.7.10. Perform hand hygiene.
- 5.7.11. Send specimens to pathology.

Figure 2: Blood Collection Equipment

Safety Vacutainer Needle	Winged Vacutainer Needle	Single use Needle Holder
		
Blood transfer device		Blood collection safety needle and vacutainer
		

6.0 Possible Complications During or Post Venepuncture

- Patient's may experience a vasovagal/syncope reaction.
- Other complications include:
 1. Nerve strike
 2. Arterial puncture
 3. Haematoma
 4. Haemolysis of the sample⁶

Related Documents

[SCGOPHCG Specimen Collection for Microbiological Examination, Infection Prevention and Control Guideline](#)

References

1. Virginia Fernandez de Pinedo Perez RN. Evidence Summary. Blood Specimen Collection: Venipuncture (Adults). The Joanna Briggs Institute EBP Database, JBI@Ovid. 2017; JBI145.
2. Sharma L. Venepuncture [Evidence based summary on the internet]. Adelaide: Joanna Briggs Institute; 2012 [updated 2012 May 08; cited 2013 July 23]. Available from: JBI Connect + level of evidence : guideline
3. Phlebotomy basics. Nursing Made Incredibly Easy! 2016;14(4):41-42.
4. PathWest Laboratory Medicine WA Pathology Policies and Procedures Manual. Standard venepuncture procedure. 2020 [cited 2020, June 6]
5. Sivapuram M S Venepuncture (Adults): Risks of Blood Specimen Collection [Evidence Based Summary on the internet].Adelaide. Joanna Briggs Institute; 2019 [cited 2020 June 6]; Available from JBI Connect + level of evidence: guideline
6. Ialongo C, Bernardini S. Phlebotomy, a bridge between laboratory and patient. Biochemia Medica.2016;26(1):17-33.
7. The Joanna Briggs Institute on Venipuncture (Adults): **Blood Specimen Collection [JBI Recommended Practice]**. **Adelaide:** November 2019 The Joanna Briggs Institute; 2019 [updated 2019 Nov 17; cited 2020 Oct]. Available from: The Joanna Briggs Institute EBP Database, JBI@Ovid.

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1: Evacuated Specimen Collection Guide

If **blood culture** sampling is indicated, this should be the **first sample** in the order of draw.

Recommended Order of Draw	Cap Colour(s)
1. Blood Cultures	
2. Sodium Citrate*	
3. Sodium Citrate (ESR tube) N.B can be collected at the end of the draw as tests performed on this tube are not compromised by other additives.	
4. ACD	
5. Serum with or without clot activator, with or without gel	
6. Rapid Serum Tube (RST) N.B. For use in Regional labs only	
7. Heparin with or without gel plasma separator	
8. EDTA with or without gel separator	
9. PTH (EDTA with gel)	
10. Trace metals EDTA If collecting for Chromium, Zinc and Cobalt, collect separately using a new luer adapter to avoid contamination	
11. Fluoride Oxalate	
12. Lithium Heparin Gas Syringe For arterial / venous blood gas collection Collect separately using a new luer adapter to avoid contamination Please consult your local laboratory for specific site requirements	

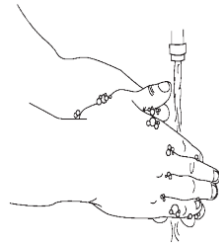
Appendix 2: Phlebotomy Troubleshooting

Problem	Cause	Signs & Symptoms	Prevention	Intervention
Failed Venepuncture	Technician factors: - Poor technique - Poor preparation - Poor vein selection Patient factors: - Poor/small veins	- No flash back in chamber - Pain - Haematoma	- Careful vein selection - Anchor vein during placement - Use heat to improve venous dilation - Use gravity to improve venous dilation - Consider practitioner skill level –seek advice prn	- Remove needle - Apply pressure to site - Seek assistance if difficulties encountered
Vasovagal/ Syncope Reaction	Fear of needles	- Light headedness - Nausea - Feeling hot and cold - Sweaty - Fainting	- Establish if patient has a fear of needles - Instil confidence in the skill of practitioner - If patient has poor vasculature only an experienced practitioner to attempt - Lay patient flat during procedure - Use heat to improve venous dilation and gravity - Consider use of topical anaesthetic	- Discontinue procedure - If patient is sitting upright, instruct them to place head between their legs - If patient faints lie flat with feet elevated - Administer oxygen and monitor vital signs - MET call if necessary
Nerve Strike	- Blind probing - Poor vein selection	- Extreme pain - Electric shock - Numbness	- Sound knowledge of anatomy - Avoid inner aspect of wrist - Palpate thoroughly - Stabilise limb - Anchor vein - Never probe blindly for a vein	- Stop procedure if patient complains of shooting or radiating pain - Inform MO if nerve damage is suspected - Document incident CIMS
Arterial Puncture	- Atypical patient anatomy - Blind probing	- Pain - Spurting bright red blood - Pulsatile vessel	- Never probe blindly - Careful vein selection - Palpate for non-pulsating vessel prior to venepuncture - Stabilise limb	- Remove needle - Apply digital pressure for 5 minutes or until haemostasis - Elevate limb - Document incident CIMS

Appendix 3: Flow diagram for best practice in venepuncture



1. Identify and prepare the patient.



2. Perform hand hygiene



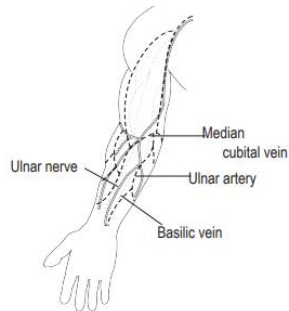
3. Assemble Venepuncture Procedure Equipment



4. Apply a tourniquet, approximately 12cm above the venepuncture site



5. Perform hand hygiene and apply well-fitting, non-sterile gloves.



6. Palpate and identify appropriate vein, checking for vein position and direction using one of the first two fingers on your non dominant hand.



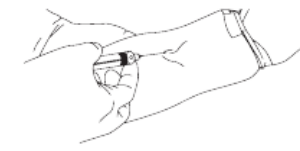
7. Disinfect the site using 2% chlorhexidine and 70% alcohol swab for 30 seconds and allow to dry completely (30 seconds).



8. Using your non dominant hand, apply traction 10-12cm's to skin below intended puncture site to anchor vein and prevent movement.



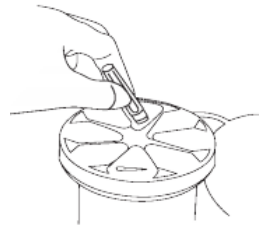
9. Enter the vein swiftly at a 15-30 degree angle.



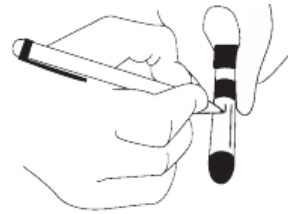
10. Once sufficient blood has been collected, release the tourniquet BEFORE withdrawing the needle.



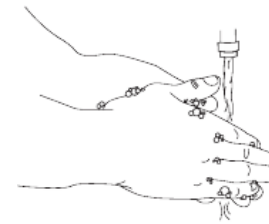
11. Withdraw the needle gently and apply pressure to wound site using a non-touch technique and apply a small sterile dressing.



12. Dispose of sharps appropriately.



13. Check the label and forms for accuracy



14. Remove gloves and place them in the general waste. Perform hand hygiene. If using soap and water, dry hands with single-use towels